ANSWER KEY & MARKING SCHEME SET A

Geography Chapter 6: **Manufacturing Industries** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** manufacturing	Secondary sector
2	**(d)** aluminium smelting	Alumina
3	**(c)** 1854	Cotton textiles
4	**(b)** Rishra	Hugli basin
5	**(a)** Both A and R are true and R is the correct explanation of A	Seasonal
6	**(b)** 4 : 2 : 1	Heavy industry
7	**(b)** Chota Nagpur plateau	Raw materials nearby
8	**(d)** Bengaluru	Electronics industry hub
9	**(a)** 1904	Construction industry
10	**(d)** joint sector industry	Jointly owned

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Industries that supply their products as raw materials to manufacture other goods, e.g. iron and steel, copper smelting, aluminium smelting.	2
12	It is the only self-reliant industry in India that covers the entire chain from raw material to the highest value-added products.	2
13	India does not have any reserves of commercially usable potash or potassium compounds in any form.	2
14	When hot water from factories and thermal plants is drained into rivers and ponds before cooling, harming aquatic life.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Modernises agriculture and reduces dependence on agricultural income by providing jobs in secondary and tertiary sectors; industrial development helps eradicate unemployment and poverty and reduces regional disparities; export of manufactured goods brings foreign exchange; transforming raw materials into high-value goods makes countries prosperous.	1+1+1
16	Proximity to jute-producing areas; inexpensive water transport supported by railways, roadways and waterways; abundant water for processing raw jute; cheap labour from West Bengal, Bihar, Odisha and UP; Kolkata provides banking, insurance and port facilities for export.	3
17	Cane produced there has a higher sucrose content; cooler climate ensures a longer crushing season; cooperatives are more successful in these states.	1+1+1
	OR: Air — undesirable gases like SO_2 and CO, dust, smoke from chemical and paper factories, brick kilns, refineries, smelting plants; Water — organic and inorganic wastes and effluents from paper, pulp, chemical, textile, dyeing, refineries, tanneries; Land — dumping of glass, chemicals, salts; Noise — machinery, generators, drills.	1+1+1
18	Public sector — owned and operated by government agencies (BHEL, SAIL); Private sector — owned by individuals or groups (TISCO, Bajaj Auto, Dabur); Joint sector — jointly by state and individuals (Oil India Ltd.); Cooperative sector — owned by producers or suppliers of raw materials/workers (sugar industry in Maharashtra, coir industry in Kerala).	3
	OR: It provides large-scale employment to weavers in their homes as a cottage industry; promotes self-reliance and swadeshi; reduces dependence on mill-made foreign cloth; uses traditional skills.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks

19	Minimise water use by reusing and recycling; rainwater harvesting; treat hot water and effluents before release — primary (mechanical: screening, grinding, flocculation, sedimentation), secondary (biological), tertiary (biological, chemical, physical with recycling); regulate overdrawing of groundwater; fit smoke stacks with electrostatic precipitators, fabric filters, scrubbers, inertial separators; use oil or gas instead of coal; silencers on generators; redesign machinery; NTPC's example — latest techniques, ash utilisation, green belts, ash pond management, ecological monitoring.	5
	OR: Ancient hand spinning and handloom; setback in colonial period; first successful mill in Mumbai (1854); boost during world wars; concentrated in cotton belt of Maharashtra and Gujarat due to raw cotton, market, transport, ports, labour, moist climate; close links with agriculture; spinning centralised in Maharashtra, Gujarat, Tamil Nadu, weaving decentralised; world-class spinning but low-quality weaving; handloom, powerloom and mills; khadi as cottage industry.	5
20	Basic industry — all other industries depend on it for machinery; used for engineering goods, construction, defence, medical, telephonic, scientific equipment; production and consumption of steel is an index of development; heavy industry — raw materials and finished goods bulky; iron ore, coking coal, limestone 4:2:1, manganese; ideally located near raw materials with efficient transport; Chota Nagpur — low-cost iron ore, high-grade raw materials in proximity, cheap labour, vast home market.	5
	OR: Aluminium — second most important metallurgical industry; light, corrosion-resistant, good conductor, malleable; plants in Odisha, WB, Kerala, UP, Chhattisgarh, Maharashtra, TN; factors — regular electricity and assured raw material. Chemical — inorganic (sulphuric acid, nitric acid, alkalies, soda ash, caustic soda) and organic (petrochemicals); its own largest consumer. Fertilizer — nitrogenous (urea), phosphatic, DAP and complex NPK; potash imported; Gujarat, TN, UP, Punjab, Kerala contribute half.	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Geography Chapter 6: **Manufacturing Industries** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** electronic	Exercise question
2	**(b)** consumer industries	Main role
3	**(d)** Bangladesh	Largest producer of raw jute
4	**(b)** gur and khandsari	Second in sugar
5	**(a)** Both A and R are true and R is the correct explanation of A	Weight-losing raw material
6	**(b)** liberalisation	New models
7	**(b)** cement	Limestone
8	**(c)** solid wastes in India	Pollution
9	**(c)** EMS (Environment Management System) 14001	Proactive approach
10	**(a)** eight times	Control of degradation

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Production of goods in large quantities after processing raw materials into more valuable products.	2
12	Agro-based industries use agricultural raw materials (cotton, jute, sugar, edible oil); mineral-based industries use minerals and metals (iron and steel, cement, aluminium, petrochemicals).	1+1
13	Basic chemicals undergo processing to produce other chemicals used for industrial application, agriculture or consumer markets.	2
14	Regular supply of electricity and an assured source of raw material (bauxite) at minimum cost.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Capital investment — small scale industry defined by maximum investment allowed on assets (at present ₹1 crore); large scale if more. Bulk — heavy industries like iron and steel (heavy raw materials and products); light industries like electrical goods (light raw materials and products).	1½+1½
16	Agro-industries raised agricultural productivity; industries depend on agriculture for raw materials; they sell irrigation pumps, fertilisers, insecticides, pesticides, PVC pipes, machines and tools to farmers; this made production efficient.	3
17	Contributes significantly to industrial production, employment generation and foreign exchange earnings; self-reliant and complete in value chain; supports farmers, boll pluckers, ginning, spinning, weaving, dyeing, designing, packaging, tailoring workers; supports chemicals, dyes, packaging and engineering industries.	3
	OR: Covers transistors, TV, telephones, cellular telecom, exchanges, radars, computers; Bengaluru the electronic capital; other centres — Mumbai, Delhi, Hyderabad, Pune, Chennai, Kolkata, Lucknow, Coimbatore; major impact on employment generation; growth in hardware and software key to success.	3
18	Organic and inorganic wastes and effluents with dyes, detergents, acids, salts, heavy metals like lead and mercury, pesticides, plastics; control — reusing and recycling water, rainwater harvesting, treating effluents in primary, secondary and tertiary stages before release.	1½+1½
	OR: Gujarat plants have suitable access to the market in the Gulf countries; cement needs bulky raw materials like limestone, silica, gypsum, coal and power, and rail transport.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Source of raw material — agro-based (cotton, jute, sugar) and mineral-based (iron and steel, cement); main role — basic/key (iron and steel) and consumer (sugar, toothpaste); capital investment — small scale (investment up to ₹1 crore) and large scale; ownership — public (BHEL, SAIL), private (TISCO, Bajaj), joint (OIL), cooperative (sugar in Maharashtra, coir in Kerala); bulk — heavy (iron and steel) and light (electrical goods).	5

	OR: Air — SO_2, CO, particulate matter, smoke from factories, kilns, refineries; toxic gas leaks (Bhopal); affects health, plants, buildings. Water — effluents from paper, chemical, textile, tanneries; heavy metals; thermal pollution; nuclear wastes. Land — dumping of glass, chemicals, packaging, salts, garbage; percolation to groundwater. Noise — machinery, generators, drills; hearing impairment, stress, blood pressure.	5
20	Jute — India largest producer of raw jute and jute goods, second exporter after Bangladesh; mills on banks of Hugli; first mill 1855 at Rishra; Partition — three-fourths of jute area went to Bangladesh; location factors. Sugar — India second in sugar, first in gur and khandsari; bulky raw material; mills in UP, Bihar, Maharashtra, Karnataka, TN, AP, Gujarat, Punjab, Haryana, MP; 60 per cent in UP and Bihar; seasonal, suited to cooperatives; shift to south and west.	5
	OR: Manufacturing transforms raw materials into higher-value goods; modernises agriculture; provides employment in secondary and tertiary sectors; reduces poverty and regional disparities; brings foreign exchange through exports; prosperity depends on diversifying manufacturing; competitiveness in a globalised world with quality at par with international standards.	5

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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